Financial Model Validation and Early Breakeven Pathways for Liberata

A Scenario-Based Analysis of Growth, Conversion, and Cost Discipline

Author

Qianshuo (Aaron) Wang

Instructor

Prof. Luyao Zhang

Product Team Supervisor

Anish Verma

Affiliation

Liberata (Product Team)

Course / Program

DKU Independent Study (Liberata Project)

Date

20260506

Financial Model Validation and Early Breakeven Pathways for Liberata: A Scenario-Based Analysis of Growth, Conversion, and Cost Discipline

Abstract

Early-stage venture models often fail not due to a lack of ambition, but because they rely on untested assumptions about adoption timing, monetization, and cost scaling. This study examines Liberata, an open-access academic publishing platform with incentivized quality controls, and evaluates whether strategic adjustments in the initial years can accelerate the breakeven timeline.

Using a scenario-based modeling framework, three counterfactual configurations are constructed to isolate distinct mechanisms: front-loaded university acquisition, accelerated pilot-to-paid conversion, and lean operational scaling. Each scenario modifies a single dimension of the baseline model, enabling controlled comparison of their effects on revenue timing, cost structure, and profitability.

The results show that early-stage performance is highly sensitive to timing-related assumptions. Front-loaded adoption increases projected revenue from $9.0 million to $13.6 million in 2028 and advances breakeven by approximately one year. Faster conversion improves monetization and early cash flow, while cost optimization reduces operating expenses from $3.03 million to $2.66 million in 2026, improving EBIT without changing revenue.

Overall, breakeven timing is driven less by long-term growth than by the interaction between revenue timing and cost discipline in the early years. This study contributes a structured framework for financial model validation and highlights the

importance of testing high-leverage assumptions in uncertain environments.

Introduction

Financial models in early-stage ventures are often presented as forward-looking forecasts, but in reality they function as structured representations of assumptions about customer adoption, pricing, cost scaling, and organizational capacity. Rather than predicting the future, such models embed hypotheses about how a business is expected to evolve. McGrath (2010) argues that in uncertain environments, financial projections should be treated as "assumption-driven experiments," where the goal is not accuracy but learning which variables matter most for performance. In this sense, model validation is fundamentally about testing the sensitivity of outcomes to changes in underlying assumptions rather than confirming a single projected trajectory.

This perspective is closely aligned with the broader literature on business models. Teece (2010) emphasizes that a business model is not merely a description of how a firm creates value, but a system that links value creation to value capture through pricing, customer segments, and organizational design. Similarly, Zott and Amit (2010) conceptualize business models as activity systems in which the configuration and timing of activities jointly determine firm performance. These frameworks imply that small changes in one part of the model, such as customer acquisition timing or monetization structure, can have disproportionate effects on financial outcomes.

In the context of Liberata, this interdependence is particularly important because the model relies heavily on early adoption by universities. Institutional markets differ significantly from consumer markets: adoption is slower, procurement processes are

formalized, and decisions are often influenced by perceived legitimacy rather than purely economic value. Suchman (1995) defines organizational legitimacy as the generalized perception that an entity's actions are appropriate within a socially constructed system of norms and beliefs. In practice, this means that early adopters can play a critical role in reducing uncertainty for later customers. Dacin, Oliver, and Roy (2007) further show that in institutional environments, affiliations and early partnerships can enhance perceived credibility, making adoption more likely.

At the same time, monetization in business-to-business contexts is not only a function of demand but also of how value is structured and communicated. Töytäri and Rajala (2015) demonstrate that firms often fail to capture value even when they create it, due to misalignment between pricing models and customer-perceived benefits. This insight is directly relevant for Liberata's pilot-to-paid transition: delaying monetization through extended free trials may increase adoption, but it can also postpone revenue capture and weaken early financial performance.

Given these dynamics, this paper focuses on a central question: which strategic levers are most effective in improving early-stage financial performance and advancing the breakeven point? To answer this, the study constructs and compares multiple scenarios that modify key assumptions in the baseline model. Scenario planning is particularly suited for this purpose. As Bowman and MacKay (2020) argue, scenarios are not predictions but structured explorations of alternative futures that help decision-makers understand the consequences of different strategic choices. The contribution of this study is therefore twofold. First, it provides a structured

validation of Liberata's financial model by identifying which assumptions drive early-stage performance. Second, it demonstrates that breakeven timing is not determined solely by growth, but by the interaction of three distinct mechanisms: customer acquisition timing, monetization timing, and cost scaling. Understanding how these mechanisms interact is critical for designing a financially viable scaling strategy.

Method

Modeling Approach

This study adopts a scenario-based financial modeling approach to evaluate how different strategic assumptions affect Liberata's early-stage financial performance. Rather than treating the model as a predictive tool, it is used as an analytical framework to test the sensitivity of outcomes to key assumptions under uncertainty. This approach is particularly appropriate for early-stage ventures, where long-term forecasts are inherently uncertain and highly dependent on a small number of critical variables, such as customer acquisition timing, monetization structure, and cost scaling. By systematically varying these assumptions, the model enables a structured evaluation of how different strategic choices influence revenue timing, cost dynamics, and breakeven outcomes.

Importantly, the model is designed as a comparative tool rather than an optimization model. The objective is not to identify a single "best" strategy, but to isolate and evaluate the relative impact of distinct strategic levers.

Alternative approaches, including predictive statistical models, optimization models, and stochastic simulations, were considered but not adopted as the primary method. Predictive models were not appropriate due to the lack of historical adoption data for Liberata. Optimization approaches were also not used, as the objective of this study is not to identify a mathematically optimal strategy, but to evaluate the relative impact of different strategic levers. Stochastic simulation could be incorporated in future research, but a deterministic framework was selected to ensure transparency and interpretability.

Model Specification and Replicability

The full financial model, including all assumptions, parameter configurations, and scenario implementations, is available at GitHub:

https://github.com/AaronW-77/Liberata-Independent-Study-Final-

The repository contains the Excel-based financial model and presentation materials, enabling full replication and verification of the results reported in this study.

Revenue is calculated as the sum of revenues across all user segments:

$$\mathbf{Revenue_t = \Sigma\ (Users_\{i,t\} \times Price_\{i,t\})}$$

where i indexes user segments (universities, institutions, government, and private sector).

Cost is modeled as the sum of key operating and capital components:

Cost_t = Salary_t + Cloud_t + Marketing_t + SG&A_t + CapEx_t

Profitability is measured using EBIT:

EBIT_t = Revenue_t − Cost_t

All model outputs are deterministic functions of input parameters, which are explicitly defined in the accompanying Excel model.

Key Model Assumptions

The model relies on several simplifying assumptions:

- Constant pricing growth over time

- Stable conversion dynamics across cohorts

- No market saturation constraints

- Cost components scale according to predefined rates

- Adoption decisions are independent across institutions

All assumptions, parameter values, and calculations are fully documented in the Excel file provided in the repository, enabling full replication and verification of results.

Scenario Construction

To isolate the impact of different strategic levers, three scenarios are constructed. Each scenario modifies a single dimension of the model while holding other variables constant, enabling controlled comparison of causal effects.

Scenario 1: Front-Loaded University Growth

Description

This scenario modifies the timing of customer acquisition by increasing the number of initial university customers and raising early-stage growth rates. It also incorporates moderate increases in marketing spending and business-side headcount to support accelerated adoption. The scenario tests whether earlier institutional uptake can shift revenue forward and improve early-stage profitability.

Parameter Changes

To model accelerated early-stage adoption, the baseline assumptions on university acquisition and growth dynamics are adjusted as follows:

- The number of initial university customers is increased from **15 to 20**
- Early-stage university growth rates (Years 3-5) are increased from **50% to 70%**
- Mid-stage growth rates (Years 6-8) are reduced from **60% to 45%**
- Late-stage growth rates (Years 9-10) are reduced from **50% to 30%**
- Marketing and outreach spending in early years is increased from **$100,000 to $120,000 annually**
- Business-side headcount (marketing and finance personnel) is expanded from **3 to 5**

Parameter Justification

The parameter adjustments in this scenario are designed to represent a front-loaded adoption path rather than an expansion of long-term market size. The increase in initial university customers from 15 to 20 reflects a moderate improvement in early penetration within a finite market. Given that there are 187 R1 institutions in

the United States (Carnegie Classification 2025), this change corresponds to an increase from approximately 8.0% to 10.7% market coverage, which remains consistent with an early-adopter scenario. Diffusion theory further supports this assumption, as adoption under uncertainty often clusters around early adopters rather than occurring uniformly (Rogers 2003).

The increase in early-stage growth from 50% to 70% represents a high-performance but still plausible SaaS growth assumption. Empirical benchmarks show substantial dispersion in early-stage growth, with median rates ranging from 25% overall to 50% for $1–5M ARR firms, and up to 100% for smaller firms (SaaS Capital 2024; High Alpha 2024). Leading firms have also achieved growth rates around 70% in strong cohorts (Bessemer Venture Partners 2023), suggesting that this assumption lies within the upper bound of observed outcomes.

The reduction of mid- and late-stage growth (60%→45% and 50%→30%) reflects the normalization of growth as markets mature. Industry benchmarks indicate that SaaS companies at later stages typically exhibit growth rates of approximately 25%–30% (SaaS Capital 2024; High Alpha 2024), supporting a transition toward more moderate expansion over time.

The increase in marketing and outreach spending from $100,000 to $120,000 aligns with benchmark spending ratios, where marketing typically accounts for around 8% of revenue in private B2B SaaS firms (SaaS Capital 2024). At an early-stage revenue level of approximately $1.5M, this adjustment reflects a proportionally consistent increase rather than aggressive overinvestment.

The increase in business-side headcount from 3 to 5 is best understood as an internally consistent managerial assumption. As early adoption and growth accelerate, additional capacity is required to support customer acquisition, onboarding, and relationship management. This adjustment ensures that operational capacity remains aligned with higher growth expectations.

Overall, these changes represent a reallocation of growth toward earlier periods while maintaining consistency with industry benchmarks. The scenario therefore tests whether accelerated early adoption can improve profitability timing without implying an unrealistically large long-term market.

Scenario 2: Accelerated Pilot-to-Paid Conversion

Description

This scenario evaluates whether improving the efficiency of the pilot-to-paid conversion process can accelerate Liberata's path to profitability by shifting revenue recognition earlier in the model timeline.

The baseline model assumes a conservative institutional adoption process, in which universities participate in extended free trials before converting to paid subscriptions. This reflects typical procurement cycles in higher education, where evaluation periods can be lengthy due to budget approval processes and administrative complexity.

In contrast, this scenario models a more efficient conversion pathway by reducing reliance on extended free trials and introducing a partially paid pilot structure. The objective is to test whether earlier monetization—without changing growth rates or

pricing—can materially improve early-stage cash flow and shorten the breakeven timeline

Parameter Changes

To isolate the effect of conversion timing, only two parameters are modified:

・The percentage of universities receiving a free trial in Years 1-2 is reduced from **100% to 50%**

・A paid pilot structure is introduced, with pilot pricing set at **$20,000** instead of a fully free trial

All other parameters are held constant.

Parameter Justification

The reduction in free trial coverage from 100% to 50% reflects a shift toward more selective and structured pilot programs. In enterprise software markets, fully unrestricted free trials are relatively uncommon for high-value B2B products. Instead, firms often rely on controlled pilot programs with defined scope, success criteria, and limited access. This is particularly relevant in institutional contexts such as universities, where procurement decisions are formalized and often require demonstrable value before full adoption.

While precise public benchmarks on the percentage of customers receiving free trials are limited, industry evidence suggests that structured pilots tend to have higher conversion rates than open-ended free trials. Enterprise SaaS practices emphasize the qualification of high-intent customers prior to granting access, rather than offering universal free trials (Töytäri and Rajala 2015). Therefore, reducing free trial coverage

to 50% reflects a more selective and resource-efficient allocation of pilot access, focusing on institutions with a higher likelihood of conversion.

The introduction of a $20,000 paid pilot is supported by pricing practices in enterprise SaaS and academic software procurement. In high-value institutional software markets, pilot programs are often not entirely free, but instead offered at a reduced or limited-scope price. This allows vendors to:

・Signal product value

・Filter serious buyers

・Recover part of implementation and support costs

Relative to the full subscription price of $200,000, a $20,000 pilot represents approximately 10% of annual contract value, which is consistent with discounted pilot pricing commonly observed in enterprise software negotiations (OpenView 2023). This level is low enough to reduce adoption friction, while high enough to establish a financial commitment from the customer.

Crucially, only these two parameters are modified in this scenario. This is a deliberate modeling choice. The purpose of Scenario 2 is not to optimize overall performance, but to isolate the causal impact of conversion timing on financial outcomes.

If additional variables such as growth rates, pricing, or cost structure were simultaneously adjusted, it would become difficult to attribute any observed improvement in profitability specifically to changes in conversion dynamics. By holding all other assumptions constant, this scenario functions as a controlled

experiment, allowing for a clear evaluation of whether earlier monetization alone can accelerate breakeven (McGrath 2010).

Scenario 3: Lean Scaling and Operational Efficiency

Description

This scenario evaluates whether improving operational efficiency and adopting a lean scaling strategy can accelerate Liberata's path to breakeven without altering revenue assumptions.

In the baseline model, cost structures—particularly headcount expansion and operating expense growth—scale relatively aggressively in anticipation of future demand. This implicitly assumes that early investment in personnel and infrastructure is necessary to support long-term growth.

In contrast, this scenario reflects a more disciplined operational approach, in which hiring and cost expansion are more closely aligned with realized demand rather than projected growth. The objective is to reduce early-stage cash burn while maintaining the same revenue trajectory, thereby isolating the impact of cost-side optimization on profitability and breakeven timing.

Parameter Changes

To model lean operational scaling, multiple cost-related parameters are adjusted across different phases:

(1) Cost Growth Rate Adjustments

SG&A annual growth rate: **20% → 12%**

Cloud infrastructure cost growth rate: **15% → 10%**

Marketing & outreach cost growth rate: **15% → 10%**

(2) Phase 1 (Early-Stage) Headcount Optimization

Back-end engineers: **6 → 4**

Marketing & finance personnel: **3 → 2**

(3) Phase 2 (Mid-Stage) Scaling Adjustments

Additional data engineers: **5 → 3**

Other business personnel: **10 → 6**

(4) Phase 3 (Late-Stage) Efficiency Adjustments

Additional data engineers: **20 → 12**

Parameter Justification

The parameter adjustments in this scenario are calibrated based on publicly available SaaS benchmarks and reflect a shift toward a more capital-efficient operating model. This is also consistent with broader service-sector cost structures, where labor and operating expenses typically account for the majority of total costs (U.S. Census Bureau 2024).

The reduction in SG&A growth from 20% to 12% aligns with industry observations that efficient SaaS firms moderate operating expense growth to approximately 10%–15% as they prioritize cost discipline and operating leverage (OpenView 2023; Bessemer Venture Partners 2023).

Similarly, cloud and marketing cost growth are reduced from 15% to 10%, consistent with observed ranges of approximately 8%–15% for infrastructure scaling and optimized customer acquisition spending (Flexera 2023; OpenView 2023). These

adjustments reflect a more disciplined approach to cost expansion, in which spending growth is aligned with operational efficiency rather than aggressive scaling.

Early-stage headcount reductions (e.g., engineers 6→4, business personnel 3→2) reflect lean startup practices, where firms minimize fixed costs prior to product–market fit. Empirical research suggests that premature scaling—particularly in hiring—is a key driver of startup failure (Startup Genome 2019). Reducing early-stage hiring therefore represents a risk-mitigating strategy aimed at preserving flexibility under uncertainty.

Mid-and late-stage hiring adjustments (e.g., 5→3, 10→6, 20→12) follow a demand-driven scaling logic, where resource expansion is aligned with realized growth rather than projected demand. Instead of assuming immediate capacity expansion, the model delays hiring to maintain operational efficiency and reduce unnecessary fixed costs. This approach emphasizes capital discipline and avoids overcommitment in the early stages of scaling.

Overall, these parameter choices are selected within empirically observed ranges and represent a coordinated shift toward lean scaling rather than arbitrary cost reductions. The scenario is therefore best interpreted as a structured test of whether improved cost discipline alone can accelerate breakeven without altering revenue assumptions.

Results

- **Scenario 1: Front-Loaded University Growth**

Under the front-loaded university growth scenario, the model exhibits a clear

improvement in early-stage financial performance relative to the baseline projections.

・**Revenue Impact**

Early-stage revenue increases substantially due to accelerated institutional adoption. Compared to the baseline:

Revenue in 2027 increases from **$3.0 million to $4.0 million (+33%)**

Revenue in 2028 increases from **$9.0 million to $13.6 million (+51%)**

Revenue in 2029 increases from **$14.2 million to $24.3 million (+71%)**

This pattern reflects a significant forward shift in revenue timing, driven by faster university acquisition and earlier monetization.

・**Cost Impact**

Despite the more aggressive growth strategy, cost increases remain moderate and controlled:

Total costs in 2027 increase from **$3.21 million to $3.43 million (approximately +7%)**

Total costs in 2028 increase from **$5.36 million to $5.60 million (approximately +4%)**

Total costs in 2029 increase from **$5.75 million to $6.02 million (approximately +5%)**

The relatively small increase in costs, compared to the much larger increase in revenue, indicates strong operating leverage in the early stages.

・**Profitability and Breakeven**

The improvement in revenue timing translates directly into earlier and stronger

profitability:

EBIT in **2027 improves from –$0.21 million to +$0.57 million**, marking an early transition to positive operating income

EBIT in **2028 increases from $3.64 million to $7.99 million** (more than doubling)

EBIT in **2029 increases from $8.42 million to $18.25 million** (over +100% growth)

As a result, breakeven is achieved in **2027**, one year earlier than the baseline model (2028).

・**Summary Insight**

Overall, the results demonstrate that front-loading university acquisition significantly improves near-term financial performance. The gains are primarily driven by earlier revenue realization rather than structural changes in long-term growth potential. The combination of accelerated revenue growth and only modest cost increases leads to a substantial improvement in early-stage profitability.

Scenario 2: Accelerated Pilot-to-Paid Conversion

Under the compressed pilot-to-paid conversion scenario, the model shows a clear improvement in early-stage financial performance driven by earlier revenue recognition, while cost projections remain unchanged relative to the baseline.

・**Revenue Impact**

The primary effect of this scenario is a forward shift in revenue timing. Compared to the baseline:

Projected revenue in 2027 increases from **$3.0 million to $4.5 million (+50%)**

Revenue in 2028 remains at **$9.0 million**, but is recognized earlier in the adoption

cycle

This reflects the shortening of the pilot period, allowing institutions to transition to paid subscriptions more quickly and generate earlier cash inflows.

・**Cost Impact**

Importantly, cost projections remain unchanged relative to the baseline across all periods. Total costs in:

2027 remain at approximately $3.21 million

2028 remain at approximately $5.36 million

This isolation of the conversion effect allows the model to capture the pure impact of monetization timing without introducing additional cost-side assumptions.

・**Profitability and Breakeven**

The earlier revenue realization leads to a substantial improvement in profitability:

EBIT in **2026 increases from -$3,025,000 in the baseline to -$1,525,000**

EBIT in **2027 increases from -$209,750 in the baseline to +$1,290,250**, representing a significant positive shift

As a result, breakeven is achieved in **2027**, approximately one year earlier than the baseline model.

・**Summary Insight**

Overall, the results demonstrate that accelerating pilot-to-paid conversion is a highly efficient lever for improving early-stage financial performance. By shifting revenue forward without increasing costs, the model generates a disproportionately large improvement in profitability, highlighting the importance of monetization

timing in institutional SaaS models.

Scenario 3: Lean Operational Scaling

Under the lean operational scaling scenario, the model shows a clear improvement in early-stage financial performance driven by reductions in operating costs, while revenue projections remain unchanged relative to the baseline.

・**Revenue Impact**

As this scenario does not modify any revenue-side assumptions, projected revenues remain identical to the baseline across all periods. This allows the model to isolate the impact of cost efficiency on financial performance.

・**Cost Impact**

The primary effect of this scenario is a reduction in early-stage operating costs. Compared to the baseline:

Total costs in **2026 decrease from $3.03 million to $2.66 million (–12%)**

Total costs in **2027 decrease from $3.21 million to $2.81 million (–13%)**

These reductions are driven by lower hiring levels, slower cost growth rates, and more disciplined scaling of operational expenses.

・**Profitability and Breakeven**

The reduction in costs leads to a meaningful improvement in profitability:

EBIT in **2027 increases from –$209,750 in the baseline to +$193,500**, marking an earlier transition to positive operating income

EBIT remains higher than the baseline in all subsequent years, reflecting sustained cost efficiency

As a result, breakeven is achieved in **2027**, approximately one year earlier than in the baseline model.

・**Summary Insight**

Overall, the results demonstrate that improving cost efficiency through lean operational scaling can significantly enhance early-stage financial performance. By reducing early-stage burn without affecting revenue growth, the model achieves earlier profitability, highlighting cost discipline as an effective lever in capital-constrained environments.

Discussion

・**Interpretation of Findings**

The results across the three scenarios consistently highlight that early-stage financial performance is highly sensitive to timing-related assumptions, particularly the timing of revenue realization and the scaling of costs. This finding aligns with prior research on business model experimentation and uncertainty in early-stage ventures.

McGrath (2010) argues that early-stage strategy should be viewed as a process of discovery, where firms iteratively test key assumptions rather than rely on static projections. In this context, the scenario-based approach used in this study reveals that breakeven timing is not primarily driven by long-term growth rates, but by how quickly firms can convert adoption into revenue and control early-stage cost structures.

The results also support insights from SaaS industry research emphasizing the

importance of monetization timing. Empirical evidence suggests that shortening sales cycles and accelerating conversion from trial to paid users can significantly improve cash flow without requiring additional customer acquisition (OpenView 2023). This is directly reflected in Scenario 2, where breakeven is achieved earlier purely through improved conversion timing.

In addition, the role of early adopters observed in Scenario 1 is consistent with diffusion of innovation theory (Rogers 2003), which highlights how initial adoption by high-credibility institutions can accelerate broader market uptake. In institutional markets such as higher education, early university customers function not only as revenue sources but also as legitimacy anchors, reducing uncertainty for subsequent adopters (Suchman 1995).

Finally, the effectiveness of cost discipline in Scenario 3 aligns with research on lean startup practices. Early-stage firms often prioritize minimizing burn rate and delaying fixed cost expansion until demand becomes more certain (Blank 2013). The results show that even without changes in revenue, cost-side adjustments alone can meaningfully accelerate breakeven.

・Limitations and Boundary Conditions

Despite these insights, the analysis is subject to several limitations. First, the model adopts a deterministic structure, assuming fixed growth and cost trajectories without incorporating stochastic variation or uncertainty modeling. In practice, adoption processes are often influenced by external shocks, such as budget cycles, procurement delays, and institutional decision-making constraints, which may

significantly affect adoption timing.

Second, the scenarios isolate individual mechanisms by holding other variables constant. While this improves internal validity, it simplifies real-world dynamics, where revenue growth, conversion timing, and cost scaling are often interdependent. For example, faster adoption may require additional investment in sales, onboarding, and support infrastructure.

Third, the model does not incorporate competitive dynamics or operational constraints. Market competition, including alternative platforms and pricing pressure, may influence adoption and monetization outcomes. At the same time, rapid scaling may introduce bottlenecks in organizational capacity, while aggressive cost reductions may affect product quality and user experience.

Fourth, the model relies on early-stage assumptions that have not yet been fully validated with real-world data. Although parameter values are calibrated using industry benchmarks, actual performance may vary due to context-specific factors not captured in the model.

Importantly, the purpose of this model is not to generate precise forecasts, but to evaluate the relative sensitivity of financial outcomes to different strategic levers. Therefore, all results should be interpreted as directional insights rather than exact predictions.

・Implications for Research and Practice

This study has several implications for both research and managerial practice. From a research perspective, the findings highlight the importance of incorporating

timing dynamics into financial modeling. Traditional models often emphasize total market size or long-term growth rates, but this analysis shows that early-stage outcomes are more strongly influenced by when revenue is realized and how quickly costs scale. Future research could extend this framework by incorporating stochastic modeling, scenario probabilities, or empirical calibration using real adoption data. From a practical perspective, the results suggest that early-stage firms should prioritize strategies that accelerate monetization rather than focusing solely on customer acquisition. In particular, reducing pilot duration, improving conversion rates, and designing pricing structures that enable earlier revenue capture can significantly improve financial sustainability.

At the same time, firms should maintain discipline in cost scaling. Lean operational strategies such as delaying hiring, optimizing cloud infrastructure, and controlling overhead growth can reduce early-stage burn and provide greater flexibility under uncertainty.

Importantly, the comparison across scenarios suggests that different levers have different efficiency profiles. Revenue timing improvements (Scenario 2) generate the largest impact with minimal additional cost, while growth acceleration (Scenario 1) offers strong upside but requires additional investment. Cost optimization (Scenario 3), while more conservative, provides a reliable path to earlier breakeven with lower execution risk.

Overall, the findings underscore that early-stage financial performance is not solely a function of scale, but of timing, sequencing, and strategic discipline in both

revenue and cost decisions.

Conclusion

This study demonstrates that early-stage financial outcomes are primarily driven by the timing and sequencing of key operational decisions, rather than by long-term growth assumptions alone. Through a structured scenario-based modeling framework, the analysis isolates three distinct mechanisms—accelerated customer acquisition, improved pilot-to-paid conversion, and lean operational scaling—and evaluates their respective impacts on revenue timing, cost structure, and profitability.

Across all scenarios, the findings consistently show that breakeven timing can be significantly advanced through relatively targeted adjustments in the early years. In particular, accelerating monetization (Scenario 2) emerges as the most capital-efficient lever, as it improves profitability without increasing costs. Front-loaded customer acquisition (Scenario 1) also delivers strong performance gains, though it requires additional resource investment. Meanwhile, cost-side discipline (Scenario 3) provides a more conservative but reliable pathway to earlier breakeven by reducing early-stage burn.

Importantly, these results highlight that early-stage financial performance is not solely a function of scale, but of how effectively firms manage the interaction between adoption timing, monetization, and cost expansion. In institutional SaaS contexts, where sales cycles are long and trust plays a critical role, strategic emphasis on early conversion and disciplined scaling can materially improve financial sustainability.

Overall, this study contributes a practical framework for financial model validation under uncertainty. By systematically stress-testing key assumptions, scenario-based analysis enables founders and decision-makers to identify high-leverage strategic adjustments and better align operational execution with financial outcomes.

Acknowledgments

I would like to express sincere gratitude to all individuals and organizations who contributed to the development of this research.

First, I would like to thank **Prof. Luyao Zhang** for providing continuous guidance throughout the research process. Prof. Zhang offered critical feedback on the structure of the financial model, helped refine the scenario-based methodology, and provided valuable insights on how to translate analytical findings into a rigorous academic argument.

I am also deeply grateful to Product Team Supervisor **Anish Verma** for his extensive support and mentorship. Anish provided not only strategic guidance on product and business modeling, but also detailed feedback on scenario design, assumption alignment, and analytical interpretation.

In addition, I would like to express special thanks to **Mac Anderson**, whose foundational work served as the basis for this research. Mac developed the initial financial model, pricing framework, and market sizing structure for Liberata, which provided the core analytical foundation upon which all scenario modifications in this study were built.

I would also like to acknowledge the Liberata Project leadership, **Han Zhang** and **Patrick Prochazka**, for their vision and leadership. Their commitment to improving the academic publishing system and their development of Liberata as an open-access platform with incentive-aligned quality control mechanisms provided the fundamental inspiration for this work.

Furthermore, I would like to thank **all members of the Liberata project team**, whose feedback, discussions, and collaborative efforts contributed to refining the assumptions, improving the realism of the scenarios, and strengthening the overall analytical framework. The Liberata Project provided the intellectual and infrastructural foundation for this study. It contributed to (a) the core research question and conceptual framing of scenario-based financial modeling in early-stage academic publishing platforms, (b) the computational infrastructure and Excel-based modeling tools used for financial projections, (c) the mentorship structure through ongoing guidance and feedback from project leaders, and (d) a collaborative research environment that supported iterative discussion, refinement of assumptions, and scenario design.

Bibliography

Blank, Steve. 2013. *The Four Steps to the Epiphany: Successful Strategies for Products that Win.* 2nd ed. K&S Ranch. https://steveblank.com/books-for-startups/

Bessemer Venture Partners. 2023. *State of the Cloud 2023.* https://www.bvp.com/atlas/state-of-the-cloud-2023/

Bowman, Gary, and R. Bradley MacKay. 2020. "Scenario Planning as Strategic Activity: A Practice-Oriented Approach." *Futures & Foresight Science* 2 (3-4): e32. https://onlinelibrary.wiley.com/doi/abs/10.1002/ffo2.32

Carnegie Classification of Institutions of Higher Education. 2025. Carnegie Classifications Update. https://carnegieclassifications.acenet.edu/

Dacin, M. Tina, Christine Oliver, and Jean-Paul Roy. 2007. "The Legitimacy of Strategic Alliances: An Institutional Perspective." *Strategic Management Journal* 28 (2): 169-187. https://doi.org/10.1002/smj.577

Flexera. 2023. State of the Cloud Report 2023. https://www.flexera.com/blog/cloud/cloud-computing-trends-flexera-2023-state-of-the-cloud-report/

High Alpha. 2024. SaaS Benchmarks Report. https://highalpha.com/blog/saas-benchmarks/

McGrath, Rita Gunther. 2010. "Business Models: A Discovery Driven Approach." *Long Range Planning* 43 (2–3): 247–261. https://doi.org/10.1016/j.lrp.2009.07.005

OpenView. 2023. 2023 SaaS Benchmarks Report. https://openviewpartners.com/2023-saas-benchmarks-report/

Rogers, Everett M. 2003. *Diffusion of Innovations.* 5th ed. New York: Free Press. https://books.google.com/books?id=9U1K5LjUOwEC

SaaS Capital. 2024. "SaaS Growth Rate Benchmarks." https://www.saas-capital.com/blog-posts/saas-growth-rate-benchmarks/

Startup Genome. 2019. *Global Startup Ecosystem Report 2019.* https://startupgenome.com/reports/global-startup-ecosystem-report-2019

Suchman, Mark C. 1995. “Managing Legitimacy: Strategic and Institutional Approaches.” *Academy of Management Review* 20 (3): 571–610. https://doi.org/10.5465/amr.1995.9508080331

Teece, David J. 2010. “Business Models, Business Strategy and Innovation.” *Long Range Planning* 43 (2–3): 172–194. https://doi.org/10.1016/j.lrp.2009.07.003

Töytäri, Pekka, and Risto Rajala. 2015. “Value-Based Selling: An Organizational Capability Perspective.” *Industrial Marketing Management* 45: 101–112. https://doi.org/10.1016/j.indmarman.2015.02.009

U.S. Census Bureau. 2024. Service Annual Survey. https://www.census.gov/programs-surveys/sas.html

Zott, Christoph, and Raphael Amit. 2010. “Business Model Design: An Activity System Perspective.” *Long Range Planning* 43 (2–3): 216–226. https://doi.org/10.1016/j.lrp.2009.07.004